Experimental Investigation of the Stability Mechanism and Emissions of a Lifted Swirl Nonpremixed Flame

Paris A. Fokaides
Chair of Combustion Technology
e-mail: Paris.Fokaides@vbt.uni-karlsruhe.de

Plamen Kasabov

Nikolaos Zarzalis

Engler-Bunte-Institute,
University of Karlsruhe (TH),
Kaiserstrasse 12,
76128 Karlsruhe, Germany

We report on the experimental investigation of a confined lifted swirl nonpremixed flame by applying a novel Airblast nozzle (Zarzalis, N., et al., 2005, Fuel Injection Apparatus, Patent No. DE 10 2005 022 772.4, EP 06 009 563.5). 3D-laser doppler anemometry, a nonintrusive, laser-based measurement technique, is adapted for the measurement of all three mean velocity components and of the six Reynolds stress components. The determination of the temperature and mixture field occurs by employing in-flame measurement techniques. Valuable information concerning the mixing procedure, the temperature distribution, the turbulence level, and the velocity field of the flame is provided. The results demonstrate that there is sufficient residence time in the precombustion area of the lifted flame in order to achieve spatial and temporal uniformity of the mixture, leading to a quasi-premixed state. It was also found that hot reaction products, carried upstream by an annular zone of reverse flow, react with fresh unburnt mixture in a re-ignition process. The determination of the flow pattern revealed the presence of an inner weak recirculation zone in the nozzle vicinity and a dominant external recirculation zone. The examination of the probability density function of the velocity measurements was also found to be a very useful tool in terms of the analysis of the turbulence structure of the flow. The bimodal distribution in the shear layer between the downstream flow and the recirculated gases yields the existence of large scale eddies. Finally, the significant reduced NO_x emissions in the lean area were also shown by means of emission measurements for elevated pressure conditions. [DOI: 10.1115/1.2749279]

Introduction

Partially premixed flames represent a class of hybrid flames established when a quantity of oxidizer is molecularly mixed with the fuel stream before entering the reaction zone, where additional oxidizer is available for complete combustion. This model of combustion can be used to exploit the advantages of both nonpremixed and premixed flames regarding operational safety, lower pollutant emissions, and flame stability. Lifted swirl flames are a representative application of partially premixed combustion, and are considered to be the breakthrough that should enable high-level reduction in NO_x emissions. Despite their importance, many aspects of lifted swirl flames have not been adequately investigated and consequently are not clearly understood. The present work is motivated by the aspects of this promising art of combustion, and it is designed to increase knowledge of the fundamental factors governing the stability of the confined lifted swirl nonpremixed flame.

Experimental Setup

The studied atomizer is an assembly based on the Airblast concept (Figs. 1) [1,2]. The applied configuration is of the prefilming type, in which the liquid is first spread out in a thin, continuous sheet by means of an annular gap (1) and then subjected to the atomizing action. Two separate concentric swirled airflows (2,3) that flow through a number of equispaced tangential ports (4,5), allow the atomizing air to impact on both sides of the fuel sheet, and ensure the thorough mixing of air and fuel. In case gaseous fuels are applied, the atomizer lip is slightly modified, in order to enable the feed of the gas through an annular gap within the primary air flow. Three different atomizers were tested for the purpose of this investigation: one for the atmospheric experiments and two for the elevated pressure experiments. The configuration for the atmospheric experiments resulted out of the experimental work of the authors within previous research projects [3]. The high swirl number of the primary swirl body (4) results in the breakdown of the vortex, whereas the construction of the secondary radial swirler (5) encounters the displacement of the flame further downstream and thus its lift-off. A very detailed description of this specific construction and the physics behind its configuration is provided in [1]. The theoretical swirl numbers of the inner as well as the outer airflow resulted according to Leuckel [4] only from the geometry data of the swirl generators, as the ratio of the angular to the axial flux momentum divided by the narrowest radius of each inner and outer channel:

$$S_{o,\mathrm{th}} = \frac{\dot{D}_o}{\dot{I}_o R} \tag{1}$$

The swirl numbers, as well as the effective areas of the nozzles, are provided to the readers in Table 1.

For the investigation under elevated pressure conditions, two similar nozzles were used, both scaled down by a geometry factor of 2. The first one was identical to the described atomizer, and the other one had a modified secondary swirl body, producing an attached swirl diffusion flame.

Two test facilities were utilized within the scope of this study: one operating under atmospheric pressure conditions and an elevated pressure combustion rig. A schematic of the atmospheric

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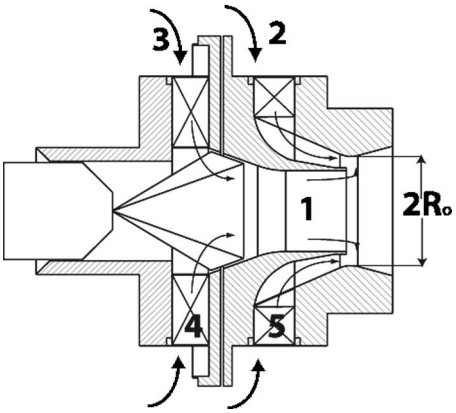

Fig. 1 Investigated Airblast atomizer [1]

test facility is seen in Fig. 2. This test stand consists of two basic segments: the nozzle holder in which the mounting and positioning of the studied nozzle occurs, and an optic accessible combustion chamber where the measurement techniques are carried out.

In order to enable the performance of field measurements, the nozzle holder can be moved axially inside the combustion chamber. Its concept allows the independent adjustment of the mass flow rate and the preheat temperature of the primary and the secondary flow. The supply of the test facility with combustion air is realized with a side channel compressor that can provide $\dot{M}=0.05$ kg/s of air with 550 mbar overpressure. The mass flow of the air is measured by variable area flow meters and the air is preheated by means of electric air preheaters. Within the nozzle

Table 1 Investigated Airblast nozzle

	$S_{o,\mathrm{th}}$	A_{eff}, mm^2	$\frac{M_i}{M_T}$
Inner	0.76	115	
Outer	≈0	225	≈0.34
Total	≈0.3	340	

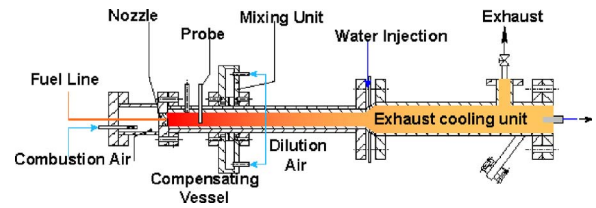

Fig. 3 Schematic setup of the pressurized combustion rig

holder, the pipe that delivers the fuel to the nozzle atomizer is water cooled in order to prevent fuel vaporization caused by the preheated air in the case in which liquid fuel is combusted.

The combustion chamber is isolated and water cooled in order to ensure the performance of reproducible measurements under adiabatic conditions. Its diameter D_{CC} is four times the throat diameter of the nozzle diffuser D_O, and its length L_{CC} was limited by an annular ring outlet at four-and-a-half times its diameter D_{CC}. This outlet geometry was chosen in order to ensure that no backflow through the exit section occurs [5]. Two properly aligned silica-glass windows enable the optical access to the combustion chamber so that the application of the LDA measurement technique could be applied. Visualization of the flame spontaneous emission by means of a CCD camera could be performed by the use of a silica glass tube chamber, whereas in this case the process is far away from the adiabatic operation point due to significant radiation heat losses (Fig. 12).

A combustion pressurized rig was applied for the measurement of the combustion emissions under elevated pressure conditions [6]. The construction of the combustor allows the realization of the RQL concept, whereas in terms of this study, this option was not utilized. The test bench is designed to host experiments up to 20 bar. It consists of five adiabatic elements, namely, the compensating vessel, the rich stage chamber, the air diluter, the lean stage chamber, and the exhaust cooling unit, all depicted in Fig. 3. Compared to the atmospheric combustor, the pressurized combustion unit is scaled down by a factor of 2. Two three-speed piston compressors are employed to provide air with a maximum mass flow rate $\dot{M}=0.125$ kg/s. Fluctuations that may occur are compensated by means of a pressurized storage vessel of 1000 l. The

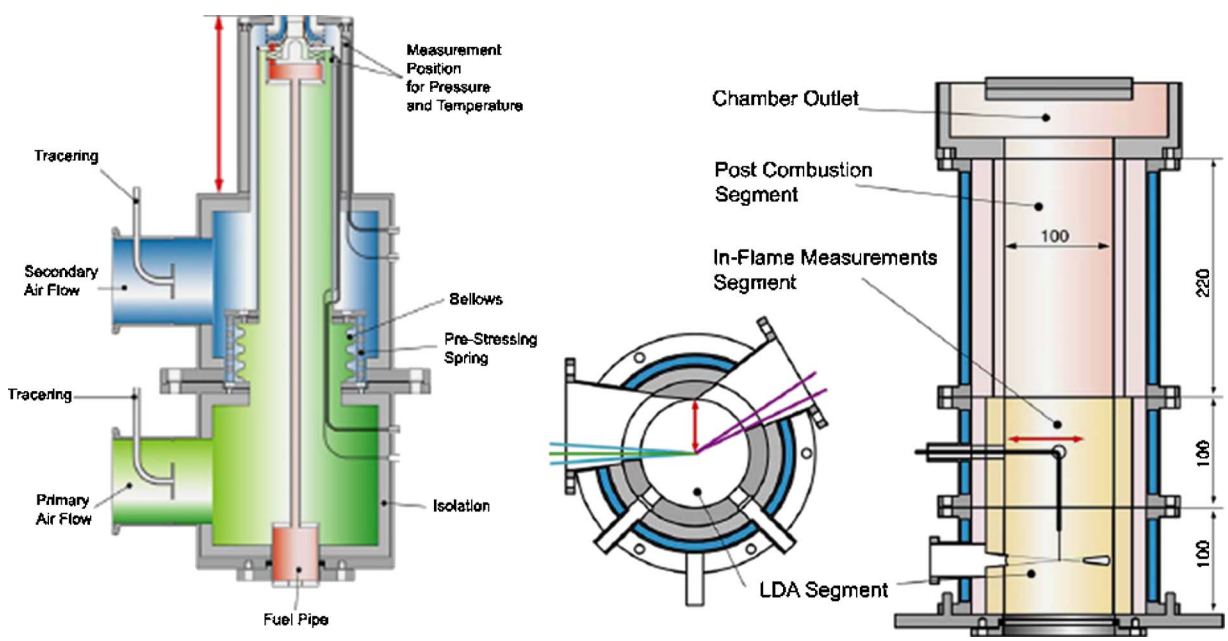

Fig. 2 Schematic setup of the atmospheric combustion chamber

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Table 2 Operating conditions for selected test cases

Atmospheric pressure experiments (methane)		
	Isothermal lifted	Reactive lifted
$T_{PR,A}$, K	323	373
P_A, bar	1	1
$\dot{M}_A$, kg/s	0.02	0.0185
U_o, m/s	42	40
Re	55,000	45,000
ΔP, %	2	2
Φ	-	0.65
Elevated pressure experiments (Jet-A1)		
	Reactive attached	Reactive lifted
$T_{PR,A}$, K	573	573
P_A, bar	7; 9	7; 9
$\dot{M}_A$, kg/s	0.02; 0.025	0.025; 0.031
U_o, m/s	45	65
Re	55,000; 70,000	85,000; 115,000
ΔP, %	3	3
Φ	0.5–1.2	0.5–1.2

air is preheated by a pressurized electrical preheater, and its mass flow is adjusted with an orifice flow meter. The mass flow of the liquid fuel (Jet-A1), supplied by a diaphragm metering pump with a vibration damper, is adjusted by a Coriolis flow meter.

The operating conditions of the performed measurements study are summarized in Table 2.

As denoted in Table 2, methane was applied for the atmospheric pressure investigation, whereas the high pressure tests were run with liquid Jet-A1 fuel, the real design for this application. The choice of the gaseous fuel under atmospheric conditions was made in order to enable the detailed investigation of the flame, by neglecting secondary parameters that could have an impact on the process and should have also been investigated, such as, for example, the two-phase flow within the flow pattern. The liquid fuel, on the other hand, does not affect the measurement techniques applied for the combustion emissions under elevated pressure conditions, and therefore was preferred in this case. Compared to the methane lifted flame, the lean blow-out limits of the Jet-A1 flame are shifted to slightly lower equivalence ratios Φ (Fig. 4), whereas the structure of the flame and its lift-off height remain similar [3].

Results and Discussion

Flow Pattern. A measurement campaign employing a commercial 3D-DANTEC LDA was conducted for the time-resolved determination of all three velocity components. The applied configuration was chosen at 30 deg off-axis forward scattering, in order to ensure a high signal-to-noise ratio and support the performance of high data rate measurements. The calculation of all three mean velocity components and of the six Reynolds stress components resulted from 20,000 to 50,000 samples measured at each grid location. As light source, a multi-line argon ion laser (COHERENT - INNOVA 90) was used at a typical operating output power of around 0.5 W. Magnesia oxide particles with diameter less than 1.2 μm, brushed off from a massive cylinder, served as tracer material for the combustion gas and air, in both air flows. Nondimensional coordinating of the system was performed based on the throat radius R_o of the diffuser. Velocities are normalized by the throat radius velocity U_O. Reynolds stresses are normalized by U_O^2 and the turbulent kinetic energy k by $\frac{1}{2}U_{O^2}$.

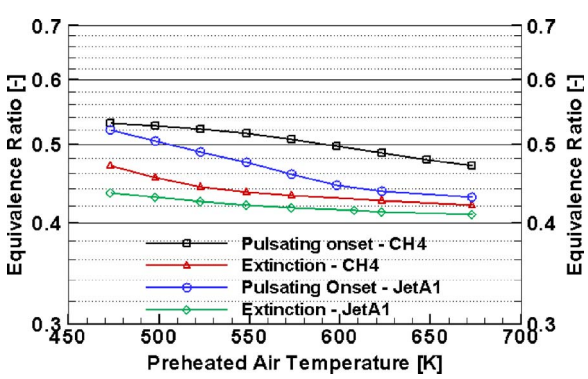

Fig. 4 Lean blow-out limits against temperature of preheated air under atmospheric conditions for methane and Jet-A1 [3]

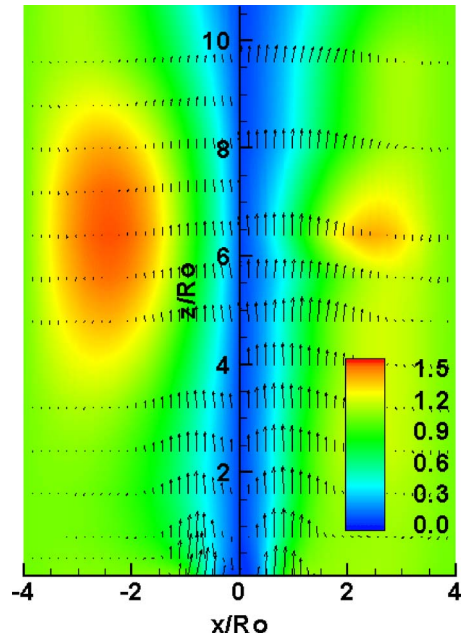

Fig. 5 Reactive (right) and isothermal flow (left): mass flow stream function Ψ and UV vector plot

Figure 5 shows the mass flow stream function Ψ contours for the isothermal and the reactive flow. In both cases, the presence of an inner weak recirculation zone in the nozzle vicinity and a dominant outer recirculation zone along the confinement is revealed. The size and the intensity of the outer recirculation zone vary in terms of the density and thus of the temperature field. According to Thring and Newby [7], the entrained air in a turbulent jet depends on the density ratio of the downstream propagating unburnt mixture to the surrounding burnout fluid. In the case of the reactive flow, the increase of this ratio results in the weakening of the outer recirculation zone.

As far as the turbulent kinetic energy k is concerned, Fig. 6 illustrates its contour for the isothermal and the reactive case. The turbulent kinetic energy per unit mass can be expressed as

$$k = \frac{1}{2}(\overline{u'}^2 + \overline{v'}^2 + \overline{w'}^2) \tag{2}$$

As expected, the elevated temperature dampens the turbulent mixing and thus postpones the completion of the procedure. Elevated turbulence levels are observed further downstream, signifying the shear layer between the upstream propagating jet and the downstream propagating hot gases. Compared to similar measurements performed with the same infrastructure and similar boundary conditions concerning an attached diffusion swirl flame [5], the reached levels of k seem to be much lower. This indicates the impact of the lower swirl number S in the case of the lifted flame.

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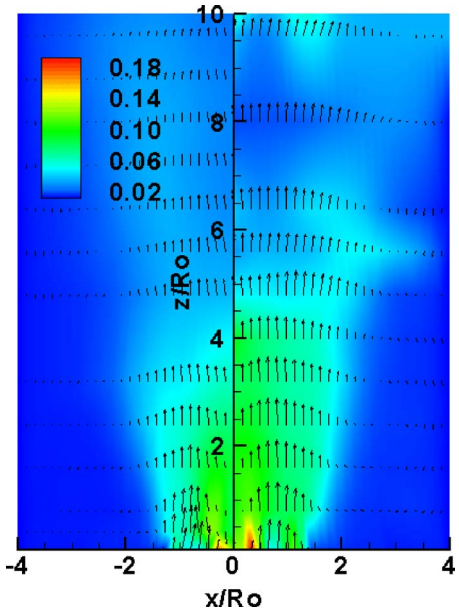

Fig. 6 Reactive (up) and isothermal flow (down): turbulent kinetic energy *k* and UV vector plot

Figure 7 compares the evolution of the mean axial velocity U under isothermal and reactive conditions. In the nozzle vicinity, negative velocities were measured on the axis in both cases, indicating the existence of a weak inner recirculation zone, which supports the mixing procedure in the precombustion zone, but does not participate in the flame stabilization process. As expected, in the reactive case, higher axial velocities are measured, whereas this difference becomes greater near and in the combustion zone. A better understanding of the impact of the combustion on the axial momentum is provided by the evolution of the local density of the axial momentum flux $\rho U^2/I_o$ (Fig. 8). A moderate increase of the axial momentum flux is observed within the first measurement planes; this effect becomes much more pronounced further downstream of the nozzle exit.

The reduced turbulence exchange in the case of the lifted flame is also confirmed from the measured turbulent shear stresses. In Fig. 9, the measured correlation $\overline{U'V'}$ for the lifted flame measurements is depicted. As expected, no deviation occurs concerning the turbulent transport across the axis, since the boundary condition of rotationally symmetric flows and thus $\overline{U'V'}(r=0)=0$ is satisfied. A quantitative comparison of $\overline{U'V'}$ for an attached and a lifted flame [5] reveals weaker turbulent exchange of axial momentum flux in radial direction in the case of the lifted flame.

The examination of the probability density function of the velocity measurements can be a very useful tool in terms of the analysis of the turbulence structure of the flow. Within the boundary layer, bimodality of the axial velocity PDFs was revealed. A bimodal distribution implies the existence, at different times, of two dominant gas directions whose temperatures and velocities vary widely and are centered at either one of the contiguous peaks. Bimodal distributions are usually located in shear layers created by mixing interfaces of fresh pockets of reactants with external recirculated hot gases. The interaction results in large

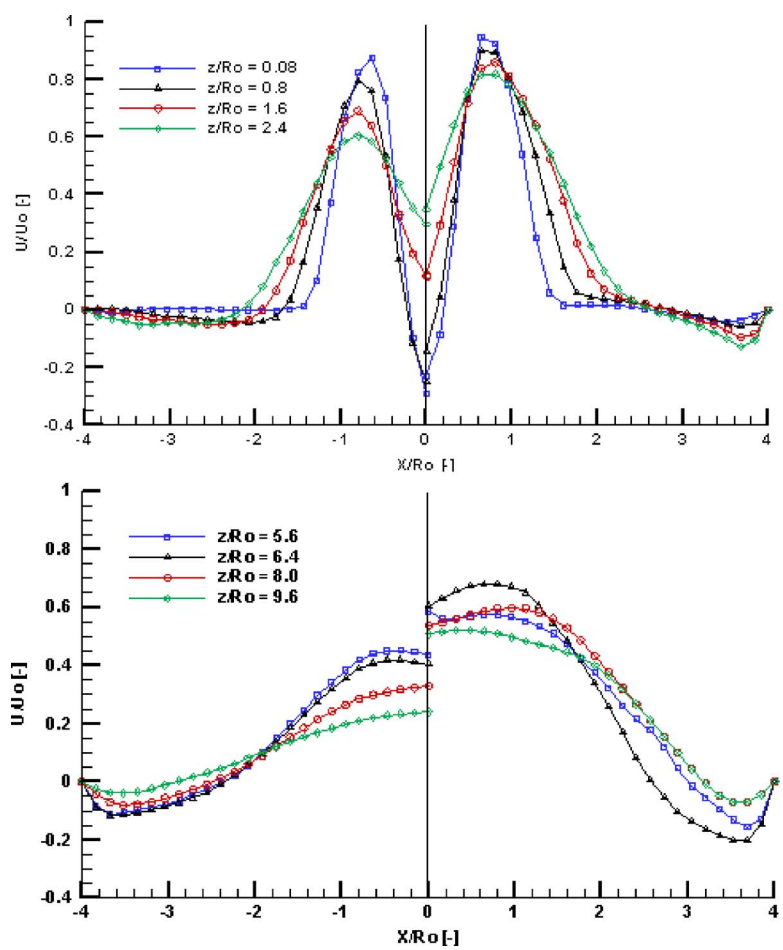

Fig. 7 Isothermal (left) and reactive flow (right): mean axial velocity $\bar{U}/U_o$

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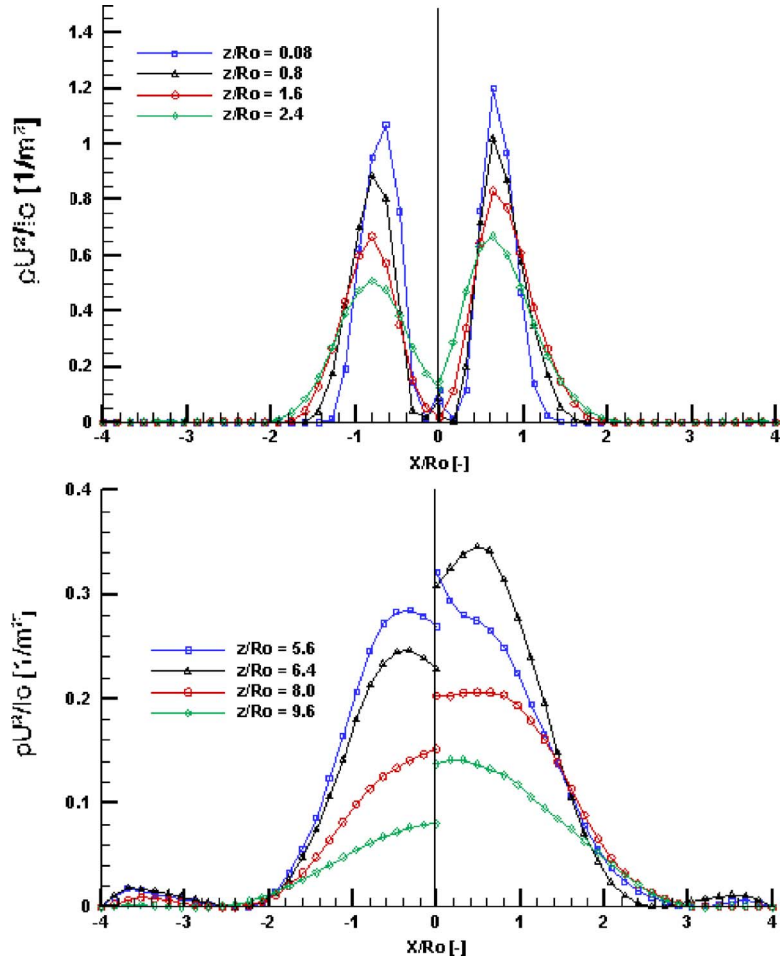

Fig. 8 Isothermal (left) and reactive flow (right): local density of axial momentum $\rho U^2/IO$

eddies of hot and cold gases existing at different times at the same location [8,9]. Figure 10 presents the evolution of the statistical distribution of the axial velocity as one moves from inside the vortex, through the shear zone into the surrounding outside gases for $z/R_o=5.6$. The distribution becomes distorted, and thus bimodal as the LDA diagnostic volume straddles the shear zone ($U=0$ m/s). This is a very interesting finding of this study, yielding the existence of large scaled eddies in the shear layer between the upstream flow and the recirculated hot gases.

Mixture Field. Measurements of multi-species concentrations in the field have been conducted in order to allow a better under-

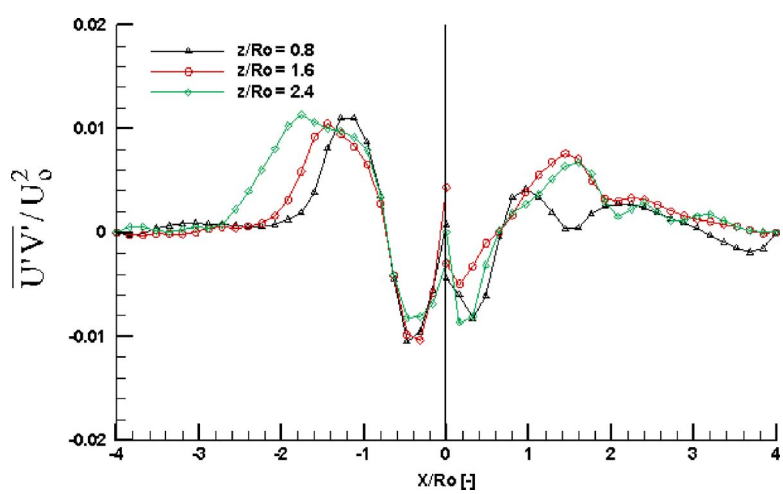

Fig. 9 Isothermal (left) and reactive flow (right): shear stress $\overline{U'V'}/U_o^2$ in nozzle vicinity

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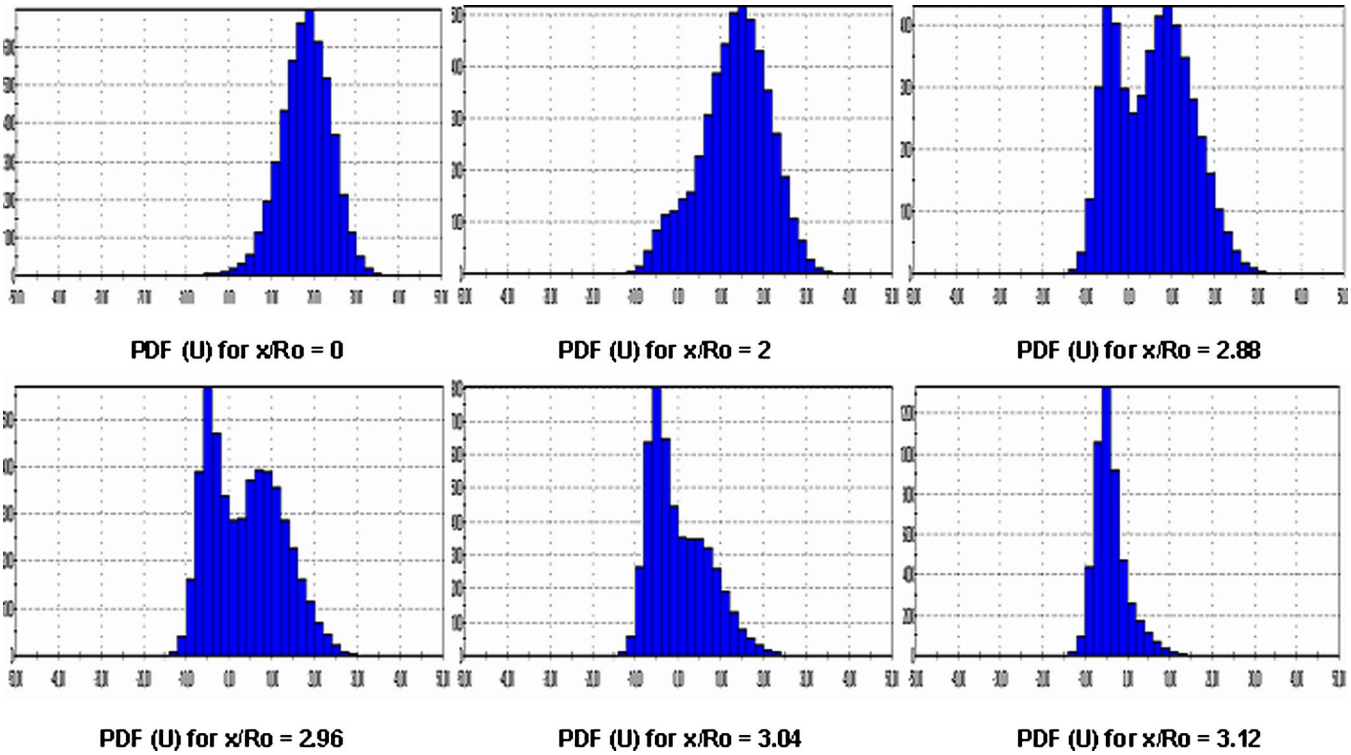

Fig. 10 Reactive flow: PDF evolution of axial velocity through shear layer for z/R_o=5.6

standing of the complex turbulence-chemistry interactions. The local concentration was subsequently analyzed by means of conventional gas analysis, based on molecular excitation processes, due to modulated infrared absorption. Two commercial analyzers (Hartmann and Braun Model URAS 14 and Magnos 14) [10] and a suction probe of 1 mm were applied.

Figure 11 presents the local equivalence ratio of the reactive field obtained by analyzed gas samples as well as the flame front, as it was detected by the carbon monoxide measurements (Fig. 15), as well as by the spontaneous emission of the flame (Fig. 12). The calculation of the local equivalence ratio from measurements of carbon monoxide, carbon dioxide, hydrocarbons, and the oxides of nitrogen uses a matrix solution of a set of ten simultaneous equations based on the combustion chemical equation of hydrocarbon fuel [11]. As this measurement yields, there is sufficient residence time below the flame base to achieve spatial and temporal uniformity of the mixture, leading to a quasi-premixed state. The results appear to be in line with previous work concerning lifted diffusion jet flames [12,13], proposing that fuel and air are fully premixed at the base of a lifted flame.

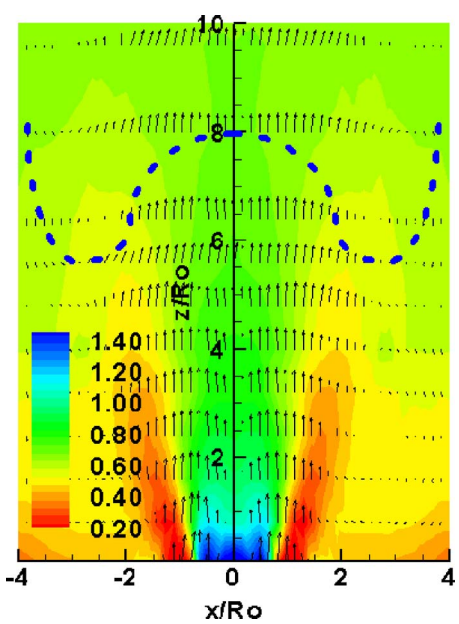

Fig. 11 Reactive flow: local equivalence ratio with flame front and UV vector plot

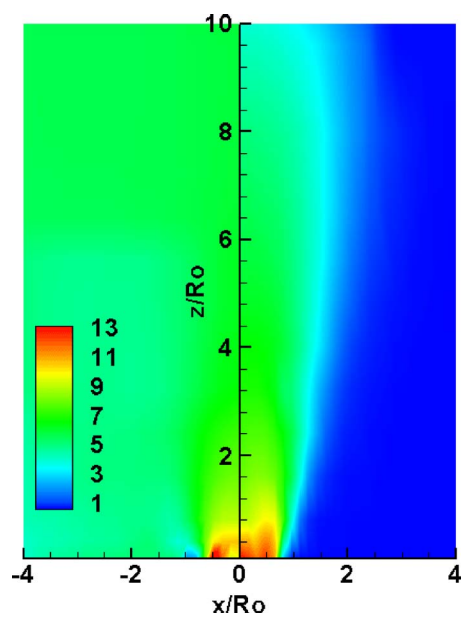

Fig. 12 Flame illumination for the investigated conditions (aperture *f*/5.6 exposure time 1/30 s)

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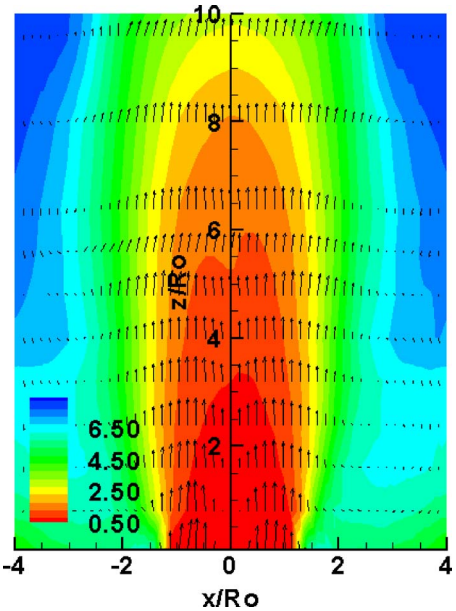

Fig. 13 Isothermal (left) and reactive flow (right): methane concentration, % vol.

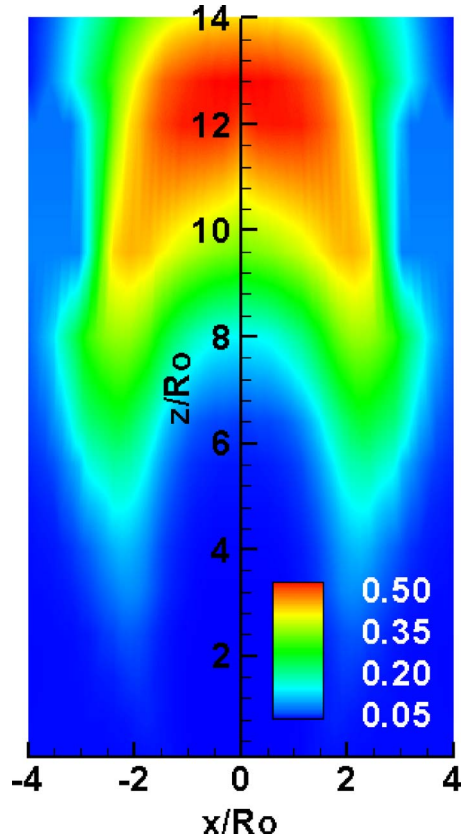

Fig. 14 Reactive flow: carbon dioxide concentration, % vol.

Compared to the isothermal conditions, the macromixing of air and fuel in the reactive case is slightly dampened. This probably occurs due to the impact of the temperature and thus of the kinematic viscosity on the diffusive mass transport. In the isothermal case, quantities of methane responding to the global adjustment of the mixture were also measured in the surrounding area of the jet, upstream from the nozzle exit, whereas in the reactive field, fuel was fully oxidized in its downstream path (Fig. 13). Due to the confinement of the flow, the evolution of an external recirculation zone resulting in the horizontal transport of methane from the upper layers to the jet's base, is to be assumed [14].

The dominance of the large recirculation zone moving the hot products upstream and parallel to the jet's boundary layer can also be confirmed from the examination of the carbon dioxide concentration in the flame. These concentration levels were found to be significant across the stabilization area, whereas in the jet core they were negligible (Fig. 14). This evidences that in the case of a confined lifted flame, hot reaction products, carried to the edges of the jet, react with fresh unburnt mixture in a re-ignition process supporting the stability of the combustion [15].

The carbon monoxide concentration is in very good agreement with the spontaneous emission captures locating the main reaction zone of the flame. The widespread region in which carbon monoxide is measured (Fig. 15) is probably a result of the effective restriction of its oxidation due to the presence of hydrocarbon species [16]. The anchor-shaped form of the reaction zone of the lifted flame in this case is very similar to the characteristic structure of the tribranchial flame [17]. However, the suspicion raised by this observation for the existence of a stabilization mechanism similar to that of the triple flame, is not supported either from the velocity field or from the local equivalence ratio.

Temperature Field. High frequency temperature measurements were performed by means of an S type (Pt/Pt-10% Rh) compensated microthermocouple probe, with wire diameter of 100 μm [18]. The raw temperature data were corrected for radiative losses in terms of a heat transfer coefficient determined from the frequency response of the compensated signal according to a function proposed by Sato and Kumura [19].

The contours of mean and rms temperatures are shown in Fig. 16. The combination of mean and fluctuating temperatures allows the location of the combustion area as well as the regions of boundary layers. The low mean temperatures near the nozzle vicinity in the jet signify that even though the mixture at this location is highly turbulent, no combustion occurs. This also occurs

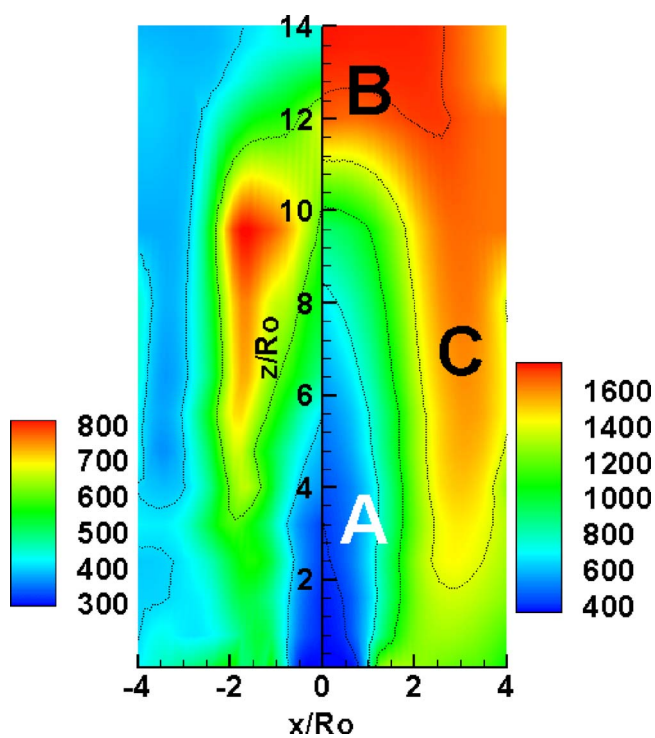

Fig. 15 Reactive flow: carbon monoxide concentration, % vol.

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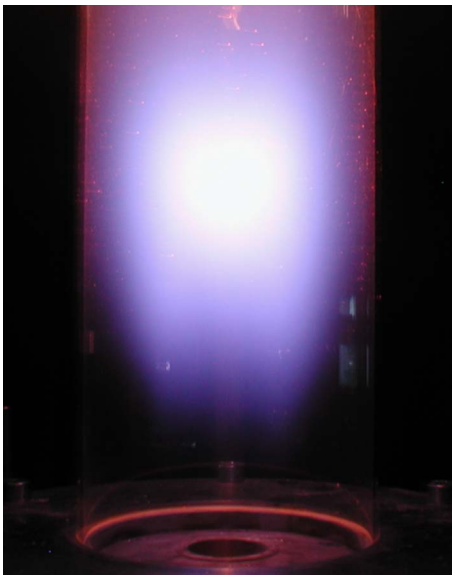

Fig. 16 Reactive flow: mean temperature (right) Trms (left) (in K) and combustion regimes (A,B,C)

due to the high axial velocities located at this regime. As the unburnt mixture approaches the combustion area, the temperature grows progressively reaching ignitable levels at $z/R_o=8$. Areas of high temperature fluctuations between the jet and the recirculated gases outside the combustion zone are due to intermittency effects caused by the fluctuating boundary layer between the unburnt jet and the combustion products.

The reaction zone is devoid of steep temperature gradients, confirming that a dominant inner recirculation zone does not exist. Temperature peaks, which usually result from the nonhomogeneous character of a diffusive flame, were also found to be avoided; i.e., quasi-premixed conditions are apparently accomplished. The highest temperatures in the field reach 80% of the adiabatic combustion temperature responding to the global stoichiometry, indicating heat losses of the system. The hottest regions themselves show only moderate levels of fluctuating temperatures, apparently due to the mixture homogeneity [20].

Another important observation concerns the spatial structure of the reaction zone in terms of an arrow-shaped front propagating upstream and parallel to the jet shear layer towards its edge. This procedure is apparently potentiated by entrainment phenomena [7], already described in this paper in terms of a dominant outer recirculation zone. The elevated temperature conditions located at the jet base are expected to have a major impact on the increase of the local laminar burning velocity and, thus, on the local turbulent burning velocity. The flammability limits of the mixture at this position are also expected to rise significantly. Acting on the assumption of the Vanquickenborne model [12], the combustion should be possible in higher flow velocities and, thus, in a lower equivalence ratio. This indicates that extended lean blow-out limits are expected to be achieved in the case of the confined swirled lifted nonpremixed flame, enabling the progress towards lean combustion.

Flame Topology. A brief recapitulation can be summed up out of the findings of this study, concerning the topology of a lifted confined swirl nonpremixed flame, in terms of the mixing procedure, the temperature distribution, the turbulence levels, and the velocity distribution within the flame. Four regions can be identified, with the precombustion area being the most important of them, concerning the conditions that have to be fulfilled in order to achieve the stabilization of the lifted flame (Fig. 16):

(a) **Precombustion area:** A weak inner recirculation zone accelerates the mixing procedure. The flow is highly turbulent and the macromixing is achieved. The temperature gradually rises from the preheating temperature of the mixture to ignitable levels. There is also an influence of the temperature field on the flux of the axial momentum.

(b) **Combustion zone:** The temperature reaches progressively its maximum levels, corresponding to 80% of the adiabatic combustion temperature. No recirculation zone is observed within the combustion region, and the reaction is devoid of temperature peaks. CO emission indicates the combustion zone, which has an arrow-shaped form. Large scale eddies exist in the shear layer between the upstream flow and the recirculated gases.

(c) **Outer recirculation area:** Part of the combusted products performs an upstream recirculation next to the downstream propagating unburnt mixture, and thus these hot reaction products react with fresh unburnt mixture in a re-ignition process, supporting the stability of the combustion. This procedure is of great importance for the maintenance and stabilization of the lifted combustion, since efforts to ignite the lifted flame in a nonconfined environment failed. Areas of high temperature fluctuations between the jet and the recirculated gases outside the combustion zone occur due to intermittency effects.

(d) **Postflame region:** The reactive species are fully oxidized resulting in the existence within the mixture of the expected combustion products of the global mixture adjustment (not labeled in Fig. 16). The temperature approaches constant levels and equals the adiabatic combustion temperature, devoid of the heat losses due to radiation and heat transfer to the chamber walls due to heat conduction and convection.

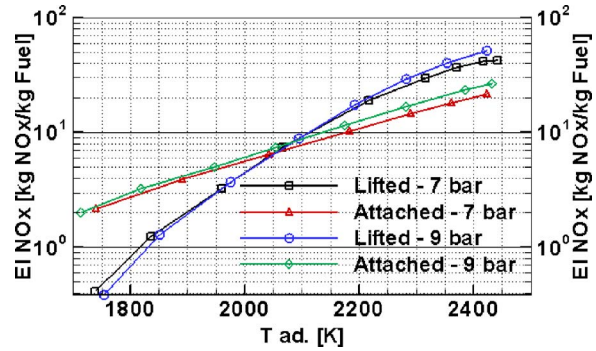

Fig. 17 EINO_x emissions (kg NO_x/kg fuel) for lifted and attached flame against adiabatic flame

NO_x Emissions. In order to approve its operability, the lifted swirl nonpremixed flame was also tested under high pressure conditions in the pressurized rig described in the experimental setup section. Within the scope of these experiments, the applied nozzle was scaled down by a factor of 0.5 and was compared to a similar Airblast configuration producing an attached swirl flame. The experiments were carried out with Jet-A1 fuel, in order to achieve near service conditions. Gas analysis was performed by means of Fourier transform infrared spectroscopy, focusing on the emissions of nitric oxide (NO_x).

The production of NO_x depends on a variety of parameters, such as the local temperature, the local mixing rates, the global flame residence time, and the radiative heat losses [21]. According to the main chemical pathway responsible for NO_x formation in combustion (the thermal NO pathway postulated by Zeldovich), temperatures above 1700 K are required in order to significantly activate the NO production [22].

NO_x emissions for the described nozzles plotted against the adiabatic flame temperature are depicted in Fig. 17. The experi-

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ments were performed for preheating temperatures of 573 K, pressures of 7 and 9 bar, and an equivalent ratio Φ range from 0.5 to 1.2. The results demonstrate that the lifted flame achieves, as expected, the required objective, namely, the reduction of the NO_x emissions in the lean area, and thus in adiabatic combustion temperatures below 2000 K. This occurs due to the premixed character of the lifted flame. The absence of an inner recirculation zone in the flame also results in a lower residence time within the hot regime of the flame and thus in lower NO_x emissions [23].

Conclusions

This study aimed to investigate a confined lifted swirl nonpremixed flame. The results demonstrated that, for the investigated flame, a significant amount of the total fuel flow was already mixed at a macrolevel upstream of the flame stabilization zone. Temperature peaks, resulting from nonhomogeneity, were found to be avoided and hence combustion with low production of the pollutant NO_x was accomplished. The combination of velocity and temperature maps correlated to the mixture field made it possible to locate the stabilization area, the recirculation zones, as well as the postflame region. The results showed a flow field devoid of a large dominant inner recirculation zone. A significant amount of exhaust gases was found upstream from the reaction zone and perimetrical from the downstream jet, forming an outer recirculation zone that apparently dominated the stabilization of the lifted swirl nonpremixed flame. This procedure is of vital importance for the maintenance and stabilization of the lifted combustion, since efforts to ignite the lifted flame in a nonconfined environment failed. The lifted swirl nonpremixed flame was also found to have significant reduced NO_x emissions and elevated pressure conditions at the lean area under, compared to a similar Airblast configuration producing an attached swirl flame.

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Nomenclature

D = diameter, m
D = angular momentum, kg m^2/s
I = axial momentum, kg m/s
L = length, m
$\dot{M}$ = mass flux, kg/s
P = power, W
R = radius, m
Re = Reynolds number
S = swirl number
T = temperature, K
U = axial velocity component, m/s
V = radial velocity component, m/s
K = turbulence kinetic energy, m^2/s^2
P = pressure, bar
X = radial coordinate, m
Z = axial coordinate, m

Greek Symbols

Δ = finite difference
Ψ = mass flow stream function
P = density, kg/m^3
Φ = equivalence ratio

Subscripts

A = air
AD = adiabatic
CC = combustion chamber
O = referencing nozzle throat
PR = preheating

Operators

$-$ = time averaged
$'$ = fluctuation

Abbreviations

CCD = charge coupled device
$EINO_x$ = emission index of nitric oxides
LDA = laser Doppler anemometry
PDF = probability density function
rms = root mean square
RQL = rich quench lean

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